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Raising a new Generation of Leaders

MTH213/MAT213: Differential Equation I

Bernoulli Equations:

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Introduction

An equation of the form

$$y' + p(x)y = q(x)y^n, \ \ n \neq 0, 1 \qquad 1$$

where $p(x)$ and $q(x)$ are functions of x or constants called a Bernoulli equation.

. Examples include are:

1. $y' + y = xy^2$
2. $\dfrac{dy}{dx} + \dfrac{y}{x} = \dfrac{y^2}{x^2}$

Bernoulli equations are non linear but can be transformed into a linear equation by substitution.

Steps to solve a Bernoulli equation

Step 1: Identify the form: Ensure the equation is in the standard Bernoulli form:

$$\frac{dy}{dx} + p(x)y = q(x)y^n.$$

Step 2: Determine n: Identify the value of the exponent n (where $n \neq 1, 0$).

Step 3: Divide by y^n: $\frac{1}{y^n}\frac{dy}{dx} + p(x)y^{1-n} = q(x)$

Step 4: Apply substitution: Substitute $v = y^{1-n}$ then $\frac{dv}{dx} = (1-n)y^{1-n-1}\frac{dy}{dx} \Rightarrow \frac{1}{1-n}\frac{dv}{dx} = \frac{1}{y^n}\frac{dy}{dx}$ into the equation.

Step 5: Rewrite the original equation in terms of v and x:

$$\frac{1}{1-n}\frac{dv}{dx} + p(x)v = q(x) \Rightarrow \frac{dv}{dx} + (1-n)p(x)z = (1-n)q(x).$$

Contd.

Step 6: Solve the linear equation: The new equation will be linear in v and can be solved using an integrating factor $(\mu(x) = e^{\int P(x)dx})$.

Step 7: Substitute back: Once you have the solution for v, substitute y^{1-n} back in to find the solution for y.

Example: Solve $y' + y = xy^2$.
Solution:
Here $n \neq 0$ *or* 1
Divide by y^2 to obtain $\frac{1}{y^2}\frac{dy}{dx} + y^{-1} = x$
Use the substitutions $v = y^{1-2} = y^{-1}$ and
$\frac{1}{1-2}\frac{dv}{dx} = \frac{1}{y^2}\frac{dy}{dx} \Rightarrow \frac{dv}{dx} = -\frac{1}{y^2}\frac{dy}{dx}$ to obtain $-\frac{dv}{dx} + v = x$

Contd.

Rearrange into a linear equation: $\dfrac{dv}{dx} - v = -x$.

Solve: The integrating factor is $\mu(x) = e^{\int -1dx} = e^{-x}$.

$\Rightarrow (ve^{-x})' = -xe^{-x}$

Integrate both sides to obtain: $ve^{-x} = -xe^{-x} + e^{-x} + C$

Substitute back $v = y^{-1}$, $y^{-1}e^{-x} = -xe^{-x} + e^{-x} + C$.

$\Rightarrow y^{-1} = -x + 1 + Ce^{-x}$ or $y = \dfrac{1}{-x + 1 + Ce^{-x}}$

Exercises

1. $\frac{dy}{dx} + \frac{1}{3}y = e^x y^4$
2. $x\frac{dy}{dx} + y = y^2 x^2 \ln x$
3. $x^2 dy + y(x+y)dx = 0$
4. $x\frac{dy}{dx} + y \ln y = xye^x$
5. $x\left(\frac{dy}{dx} + y\right) = 1 - y$